

Double and Iterated Integrals

MATH 145 – Applied Calculus

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Integrating $f(x, y)$

How do we integrate a function $f(x, y)$?

How is the integral $\int_a^b f(x) dx$ defined and evaluated?

Can we extend the ideas of Riemann sums to multivariable functions?

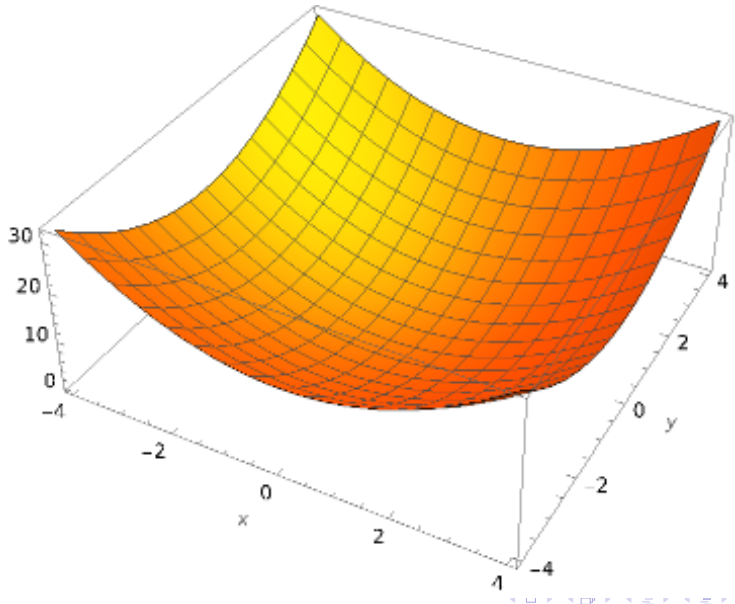
Approximating the Integral of $f(x, y)$

In the one variable case, we took intervals of the number line and estimated the area under the curve as the sum of rectangles.

In the two variable case, let's take intervals of both axes and use the rectangles to make rectangular prisms to estimate the volume under the surface.

You can expand this idea to n -dimensional functions, but we won't touch on that.

Approximating the Integral of $f(x, y)$



Defining the Integral of $f(x, y)$

Double Riemann Sum

Let f be a continuous function on a rectangle v . A **double Riemann sum** for f over the rectangle R is

$$\sum_{j=1}^n \sum_{i=1}^m (f(x_{ij}^*, y_{ij}^*) \times \delta A) .$$

Double Integral

If f is continuous over R , then the **double integral** of f over R is

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m (f(x_{ij}^*, y_{ij}^*) \times \delta A) .$$

Properties of the Double Integral

Let f and g be continuous functions on a rectangle

$R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$, and let k be a constant.

Then:

1.
$$\iint_R (f(x, y) + g(x, y)) \, dA = \iint_R f(x, y) \, dA + \iint_R g(x, y) \, dA.$$
2.
$$\iint_R kf(x, y) \, dA = k \iint_R f(x, y) \, dA.$$
3. If $f(x, y) \geq g(x, y)$ on R , then
$$\iint_R f(x, y) \, dA \geq \iint_R g(x, y) \, dA.$$

Evaluating Double Integrals

Fubini's Theorem

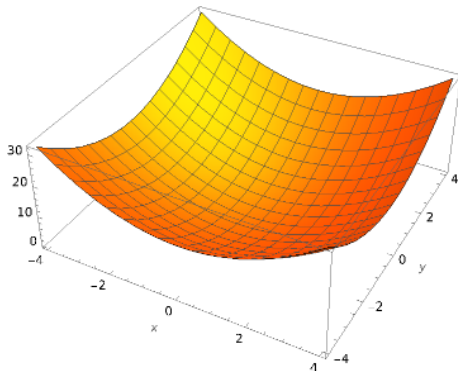
If $f = f(x, y)$ is a continuous function on a rectangle $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx.$$

$$\int \int_R lol = \iint_R lol$$

Example: Evaluating Double Integrals

Example: Find the *volume* under the graph of $f(x, y) = x^2 + y^2$ over the rectangle $-2 \leq x \leq 2$, $-2 \leq y \leq 2$.



Example: Evaluating Double Integrals

Example: Let $f(x, y) = 25 - x^2 - y^2$ on the rectangular domain $R = [-3, 3] \times [-4, 4]$. Find $\iint_R f(x, y) dA$.

Example: Find $\iint_R \sin(x) \cos(y) dA$ for $R = \{(x, y) : 0 \leq x \leq \frac{\pi}{3}, 0 \leq y \leq \frac{\pi}{4}\}$